

Comprehensive Windows VM Migration Guide

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This guide provides step-by-step instructions for migrating Windows VMs from VMware, Hyper-V, or other hypervisors to KVM using hyper2kvm with full enterprise feature support.

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Overview

hyper2kvm supports comprehensive Windows VM migration with enterprise features:

- **Automated driver injection:** VirtIO drivers for storage, network, and peripherals
- **License preservation:** Extract and reactivate Windows licenses (OEM, Retail, MAK, KMS)

- **Active Directory support:** Domain membership detection and rejoin automation
- **Application compatibility:** Detect hardware-dependent apps, license services, SQL Server
- **Performance optimization:** VirtIO balloon, TRIM/discard, MSI interrupts, Hyper-V cleanup

Supported Windows Versions: - Windows Server 2008 R2 through 2025 - Windows 7 through 11 - Both x64 and x86 (32-bit)

Prerequisites

Software Requirements

1. **hyper2kvm** v0.5.0 or later
2. **VirtIO drivers ISO** (latest from [Fedora VirtIO](#))
3. **Source VM files** (VMDK, VHDX, VHD, or OVA)
4. **KVM hypervisor** (QEMU/KVM, libvirt)

Access Requirements

- **Read access** to source VM files
- **Write access** to target storage location
- **Network connectivity** (for KMS activation if applicable)
- **Domain admin credentials** (for AD operations, if applicable)

Resource Requirements

- **Disk space:** 2x source VM size (for conversion)
 - **RAM:** 2GB minimum, 4GB recommended
 - **CPU:** Multi-core recommended for faster conversion
-

Pre-Migration Planning

1. Inventory Source VM

Document the source VM configuration:

```
# For VMware VMs
vmware-vdiskmanager -R source.vmdk

# For Hyper-V VMs
Get-VHD -Path C:\VMs\source.vhdx | Select-Object *

# Using hyper2kvm inspect
hyper2kvm inspect source.vmdk
```

Document: - Operating system version - Domain membership status - Installed applications (especially hardware-dependent ones) - License type (OEM, Retail, MAK, KMS) - Network configuration - Storage configuration (basic, LVM, encrypted)

2. Prepare VirtIO Drivers

Download and extract VirtIO drivers:

```
# Download latest VirtIO ISO
wget https://fedorapeople.org/groups/virt/virtio-win/direct-downloads/stable-virtio/virtio-win.iso

# Extract drivers
mkdir -p /opt/virtio-drivers
hyper2kvm extract-drivers virtio-win.iso /opt/virtio-drivers
```

3. Create Configuration File

Create a YAML configuration file for your migration:

```
# windows-migration.yaml
command: local
vmdk: /data/vms/windows-server.vmdk
to_output: /data/kvm/windows-server.qcow2

windows:
  # VirtIO driver injection
  virtio_drivers_path: /opt/virtio-drivers
```

```

# License management
license:
  extract: true
  reactivate: true
  kms_server: kms.example.com # Optional: Override KMS server

# Active Directory
activedirectory:
  extract_domain_info: true
  rejoin_method: unattended # manual, credential, or unattended
  ou_path: "OU=Servers,DC=example,DC=com" # Optional

# Application compatibility scanning
appcompat:
  detect_hardware_apps: true
  detect_license_services: true
  detect_sql_server: true
  generate_report: true

# Performance optimization
performance:
  balloon:
    enable: true
    memory_stats_interval: 10
    free_page_reporting: true
  trim:
    enable: true
    schedule_optimization: true
  msi:
    enable: true
    devices:
      - viostor
      - netkvm
  hyperv:
    cleanup: true # Auto-detect Hyper-V and cleanup

```

Basic Migration Workflow

Step 1: Run Migration

```

# Using configuration file
hyper2kvm migrate --config windows-migration.yaml

```

```
# Or inline for simple migrations
hyper2kvm convert \
  --input /data/vms/windows-server.vmdk \
  --output /data/kvm/windows-server.qcow2 \
  --virtio-drivers /opt/virtio-drivers \
  --windows-optimize
```

Step 2: Monitor Progress

```
# Check conversion progress
tail -f /var/log/hyper2kvm/migration.log

# Monitor resource usage
watch -n 1 'df -h /data/kvm && free -h'
```

Step 3: Create KVM VM

After successful conversion, create the KVM VM:

```
# Using virt-install
virt-install \
  --name windows-server \
  --memory 4096 \
  --vcpus 2 \
  --disk path=/data/kvm/windows-server.qcow2,bus=virtio \
  --network network=default,model=virtio \
  --graphics vnc \
  --os-variant win2k19 \
  --import

# Or using virsh with XML
virsh define windows-server.xml
virsh start windows-server
```

Step 4: First Boot Verification

Connect to the VM console and verify:

1. **Windows boots successfully**
2. **VirtIO drivers loaded** (check Device Manager)
3. **Network connectivity** established
4. **Disk performance** acceptable

Enterprise Features

License Management

Automatic License Detection

hyper2kvm automatically extracts Windows license information during migration:

```
windows:
  license:
    extract: true # Extract license info from registry
```

Extracted information: - Product key (Windows 7-11, Server 2008 R2-2025) - License type (OEM, Retail, MAK, KMS, Volume) - Product name and edition - Installation ID

License Reactivation

Configure automatic reactivation:

```
windows:
  license:
    extract: true
    reactivate: true
    kms_server: kms.company.com # Optional: Override detected KMS
    kms_port: 1688 # Optional: Custom KMS port
```

Reactivation methods by license type:

License Type	Method	Notes
OEM	Requires phone activation	Original key may not work on KVM
Retail	Automatic online activation	Usually works without intervention
MAK	Automatic online activation	Decrements activation count
KMS		

License Type	Method	Notes
	Automatic KMS activation	Requires network connectivity to KMS
Volume	Manual intervention may be required	Contact your volume license administrator

Post-Migration License Activation

After first boot, check activation status:

```
# Check activation status
slmgr.vbs /dli
slmgr.vbs /dlv

# If reactivation failed, run manually
C:\hyper2kvm\license\reactivate.ps1

# For KMS clients
slmgr.vbs /ato
```

License Report

Review the extracted license information:

```
# View license report (created during migration)
cat /data/kvm/windows-server-license-report.json
```

Example report:

```
{
  "product_name": "Windows Server 2019 Standard",
  "license_type": "Volume",
  "partial_product_key": "XXXXX-XXXXX-XXXXX-XXXXX-AB123",
  "kms_server": "kms.company.com:1688",
  "activation_required": true
}
```

Active Directory Integration

Domain Membership Detection

Automatically detect domain membership:

```
windows:
  activedirectory:
    extract_domain_info: true
```

Extracted information: - Domain name (NetBIOS and DNS) - Computer account name - Last used domain controller - Workgroup vs. domain status

Domain Rejoin Methods

Choose the appropriate rejoin method:

1. Manual Rejoin (Recommended for Testing)

```
windows:
  activedirectory:
    rejoin_method: manual
```

Steps after first boot: 1. Review instructions:

C:\hyper2kvm\activedirectory\domain-rejoin.ps1 2. Manually unjoin/rejoin using GUI or PowerShell 3. Reboot to complete

Best for: Test environments, single VM migrations

2. Credential-Based Rejoin

```
windows:
  activedirectory:
    rejoin_method: credential
    domain: example.com
    ou_path: "OU=Servers,DC=example,DC=com"
```

Steps after first boot: 1. Run: C:\hyper2kvm\activedirectory\domain-rejoin.ps1

2. Enter domain admin credentials when prompted 3. System will automatically rejoin and reboot

Best for: Interactive migrations with admin supervision

3. Unattended Rejoin (Recommended for Production)

```
windows:
  activedirectory:
    rejoin_method: unattended
    domain: example.com
    ou_path: "OU=Servers,DC=example,DC=com"
    unattended_join_file: /path/to/djoin-blob.txt
```

Preparation steps:

```
# On domain controller or admin workstation
# Pre-create computer account and generate join blob

djoin.exe /provision `
  /domain example.com `
  /machine NEW-SERVER-NAME `
  /savefile C:\djoin-blob.txt `
  /printblob

# Copy djoin-blob.txt to hyper2kvm host
```

Benefits: - No credentials stored in VM - Atomic domain join operation - Suitable for automation and templates

Best for: Production environments, mass migrations

AD Computer Object Cleanup

Remove stale computer objects from Active Directory:

```
windows:
  activedirectory:
    cleanup_old_computer: true
    old_computer_name: "OLD-SERVER-NAME"
```

Manual cleanup script is also generated:

```
# On domain controller or admin workstation
C:\hyper2kvm\activedirectory\ad-cleanup.ps1
```

This prevents conflicts when the migrated VM has a different computer name.

Application Compatibility

Compatibility Scanning

Enable comprehensive compatibility scanning:

```
windows:
  appcompat:
    detect_hardware_apps: true
    detect_license_services: true
    detect_dongle_drivers: true
    detect_sql_server: true
    generate_report: true
```

Hardware-Dependent Applications

Detects applications with known hardware dependencies:

- **Autodesk products** (AutoCAD, Maya, 3ds Max)
- **Adobe Creative Cloud** (Photoshop, Illustrator, Premiere Pro)
- **CAD/CAM software** (SolidWorks, CATIA, Mastercam)
- **Engineering tools** (MATLAB, LabVIEW)
- **Security software** (antivirus, encryption tools)

Report output:

```
{
  "hardware_apps": [
    {
      "name": "AutoCAD 2023",
      "vendor": "Autodesk",
      "risk_level": "HIGH",
      "reason": "Hardware-locked licensing (FlexLM)",
      "recommendation": "Contact vendor for license transfer"
    }
  ]
}
```

```
]
}
```

License Manager Services

Detects license services that may need reconfiguration:

- FlexLM (FlexNet Publisher)
- Sentinel HASP/LDK
- CodeMeter
- WIBU-Systems
- SafeNet Sentinel

Hardware Dongle Drivers

Detects USB dongle drivers:

- Sentinel HASP
- WIBU CodeMeter
- SafeNet eToken
- Gemalto

Mitigation: Ensure dongle is attached to KVM host and USB passthrough configured.

SQL Server Detection

Automatically detects SQL Server instances:

```
windows:
  appcompat:
    detect_sql_server: true
    generate_sql_script: true
```

Detected information: - Instance name and version - Edition (Express, Standard, Enterprise) - Installation path - Service accounts - Network configuration

Generated reconfiguration script:

```
# C:\hyper2kvm\appcompat\sql-reconfigure.sql
```

SQL Server may require: 1. Server name update (if hostname changed) 2. Linked server reconfiguration 3. Replication reinitialization 4. Reporting Services reconfiguration 5. License reactivation

Compatibility Report

Review the generated compatibility report:

```
# JSON format
cat /data/kvm/windows-server-compatibility.json

# Markdown format (human-readable)
cat /data/kvm/windows-server-compatibility.md
```

Report sections: - Executive summary (risk levels, findings count) - Hardware-dependent applications - License manager services - Hardware dongle drivers - SQL Server instances - Recommended actions

Performance Optimization

VirtIO Balloon Driver

Configure memory ballooning:

```
windows:
  performance:
    balloon:
      enable: true
      memory_stats_interval: 10 # seconds
      free_page_reporting: true # Windows 10 1809+, Server 2019+
```

Benefits: - Dynamic memory management - Memory overcommitment support - Improved host memory utilization

Verification:

```
# Check balloon service status
Get-Service balloon
```

```
# Verify registry settings
Get-ItemProperty "HKLM:\SYSTEM\CurrentControlSet\Services\balloon\Parameters"
```

TRIM/Discard for SSDs

Enable TRIM support:

```
windows:
  performance:
    trim:
      enable: true
      schedule_optimization: true # Weekly scheduled task
```

Benefits: - Improved SSD performance - Extended SSD lifespan - Reduced write amplification

Verification:

```
# Check TRIM status
fsutil behavior query DisableDeleteNotification

# Expected output: DisableDeleteNotification = 0 (enabled)

# Manually run optimization
defrag C: /L /O
```

MSI Interrupts

Enable Message Signaled Interrupts:

```
windows:
  performance:
    msi:
      enable: true
      devices:
        - viostor # VirtIO storage
        - netkvm  # VirtIO network
```

Benefits: - ~20% network throughput improvement - Reduced interrupt latency - Better CPU utilization

Verification:

```
# Run verification script
C:\hyper2kvm\performance\msi-verify.ps1

# Check registry manually
Get-ItemProperty "HKLM:\SYSTEM\CurrentControlSet\Services\netkvm\Parameters\InterruptManagement\MessageSignaledInterruptProperties" -Name "MSIInterruptFeatureEnabled"
```

Hyper-V Enlightenments Cleanup

Remove Hyper-V components when migrating from Hyper-V:

```
windows:
  performance:
    hyperv:
      cleanup: true # Auto-detect and cleanup
      force: false # Force cleanup without detection
```

Removed services (12 total): - Hyper-V Virtual Machine Bus (vmbus) - Hyper-V Guest Service Interface - Hyper-V Heartbeat Service - Hyper-V Data Exchange Service - Hyper-V Time Synchronization Service - And 7 more integration services

Verification:

```
# Check for remaining Hyper-V services
Get-Service | Where-Object {$_.Name -like "vm*" -or $_.Name -like "hv*"}

# Should return no running services
```

Post-Migration Tasks

1. Initial Boot Validation

Checklist: - [] Windows boots successfully - [] No driver missing warnings - [] Network adapters visible in Device Manager - [] Disk I/O performance acceptable - [] Event Viewer shows no critical errors

2. License Activation

```
# Check activation status
slmgr.vbs /dli

# If not activated, run reactivation script
C:\hyper2kvm\license\reactivate.ps1

# Verify activation
slmgr.vbs /xpr
```

3. Domain Rejoin

```
# For manual/credential rejoin
C:\hyper2kvm\activedirectory\domain-rejoin.ps1

# For unattended rejoin (automatic on first boot)
# Verify domain membership
Get-CimInstance -ClassName Win32_ComputerSystem | Select-Object Domain,
    PartOfDomain
```

4. Active Directory Cleanup

On domain controller or admin workstation:

```
# Remove old computer object
C:\hyper2kvm\activedirectory\ad-cleanup.ps1

# Or manually via GUI
dsa.msc # Active Directory Users and Computers
```

5. Application Validation

Review compatibility report and take action:

```
# Review compatibility report
cat windows-server-compatibility.md
```

Action items: 1. **Hardware-dependent apps:** Contact vendors for license transfer 2.

License services: Verify license server connectivity 3. **Hardware dongles:** Attach dongle via USB passthrough 4. **SQL Server:** Run reconfiguration script

```
# Run SQL Server reconfiguration
sqlcmd -S localhost -i C:\hyper2kvm\appcompat\sql-reconfigure.sql
```

6. Performance Verification

Run verification scripts:

```
# Verify all performance optimizations
C:\hyper2kvm\performance\balloon-verify.ps1
C:\hyper2kvm\performance\trim-verify.ps1
C:\hyper2kvm\performance\msi-verify.ps1
```

7. Network Configuration

Verify network settings:

```
# Check IP configuration
Get-NetIPAddress
Get-NetIPConfiguration

# Verify DNS and gateway
Test-NetConnection -ComputerName 8.8.8.8
nslookup example.com
```

8. Backup VM

Create a backup of the working VM:

```
# Create snapshot
virsh snapshot-create-as windows-server \
  post-migration-working \
  "Working state after migration and validation"

# Or clone VM
virt-clone \
  --original windows-server \
  --name windows-server-backup \
  --file /data/kvm/windows-server-backup.qcow2
```

Common Scenarios

Scenario 1: Domain-Joined Database Server

Requirements: - Windows Server 2019 with SQL Server 2019 - Domain member (example.com) - MAK license - SSD storage

Configuration:

```
command: local
vmdk: /data/vms/db-server.vmdk
to_output: /data/kvm/db-server.qcow2

windows:
  virtio_drivers_path: /opt/virtio-drivers

  license:
    extract: true
    reactivate: true

  activedirectory:
    extract_domain_info: true
    rejoin_method: unattended
    domain: example.com
    ou_path: "OU=Database Servers,OU=Servers,DC=example,DC=com"
    unattended_join_file: /data/djoin-db-server.txt

  appcompat:
    detect_sql_server: true
    generate_sql_script: true

  performance:
    balloon:
      enable: true
    trim:
      enable: true
      schedule_optimization: true
    msi:
      enable: true
```

Post-migration: 1. Verify domain rejoin successful 2. Run SQL Server reconfiguration script 3. Update SQL Server service account if needed 4. Repoint application connection strings 5. Verify database connectivity and performance

Scenario 2: Standalone Hyper-V Workstation

Requirements: - Windows 10 Professional - Migrating from Hyper-V - Retail license - Desktop applications (Office, Adobe Creative Cloud)

Configuration:

```
command: local
vhdx: /data/vms/workstation.vhdx
to_output: /data/kvm/workstation.qcow2

windows:
  virtio_drivers_path: /opt/virtio-drivers

license:
  extract: true
  reactivate: true

appcompat:
  detect_hardware_apps: true
  detect_license_services: true

performance:
  balloon:
    enable: true
    free_page_reporting: true
  trim:
    enable: true
  msi:
    enable: true
  hyperv:
    cleanup: true # Remove Hyper-V components
```

Post-migration: 1. Verify Hyper-V services disabled 2. Check license activation (may require phone activation for Retail) 3. Verify Adobe Creative Cloud license (may need reactivation) 4. Test application functionality

Scenario 3: KMS-Activated Application Server

Requirements: - Windows Server 2022 - KMS activation - Enterprise applications with FlexLM licensing - Domain member

Configuration:

```

command: local
vmdk: /data/vms/app-server.vmdk
to_output: /data/kvm/app-server.qcow2

windows:
  virtio_drivers_path: /opt/virtio-drivers

  license:
    extract: true
    reactivate: true
    kms_server: kms.company.com
    kms_port: 1688

  activedirectory:
    extract_domain_info: true
    rejoin_method: credential
    domain: company.com

  appcompat:
    detect_hardware_apps: true
    detect_license_services: true
    generate_report: true

  performance:
    balloon:
      enable: true
    trim:
      enable: true
    msi:
      enable: true

```

Post-migration: 1. Verify KMS activation (should be automatic) 2. Contact software vendors for license transfer (FlexLM) 3. Reconfigure license server settings in applications 4. Verify network connectivity to license servers

Scenario 4: Multi-VM Batch Migration

For migrating multiple Windows VMs:

```

#!/bin/bash
# batch-migrate.sh

VMs=(

```

```

"web-server-01:example.com:Web Servers"
"web-server-02:example.com:Web Servers"
"app-server-01:example.com:App Servers"
)

for vm in "${VMs[@]}; do
    IFS=':' read -r name domain ou <<< "$vm"

    echo "Migrating $name..."

    cat > "${name}-config.yaml" <<EOF
command: local
vmdk: /data/vms/${name}.vmdk
to_output: /data/kvm/${name}.qcow2

windows:
  virtio_drivers_path: /opt/virtio-drivers
  license:
    extract: true
    reactivate: true
  activedirectory:
    extract_domain_info: true
    rejoin_method: unattended
    domain: ${domain}
    ou_path: "OU=${ou},DC=example,DC=com"
    unattended_join_file: /data/djoin-${name}.txt
  performance:
    balloon:
      enable: true
    trim:
      enable: true
    msi:
      enable: true
EOF

    hyper2kvm migrate --config "${name}-config.yaml"
done

```

Troubleshooting

Boot Failures

Symptom: VM fails to boot, BSOD 0x0000007B

Cause: VirtIO storage driver not properly injected

Solution:

```
# Re-run migration with forced driver injection
hyper2kvm convert \
  --input source.vmdk \
  --output target.qcow2 \
  --virtio-drivers /opt/virtio-drivers \
  --force-virtio-injection
```

License Activation Failures

Symptom: License reactivation script fails

Causes & Solutions:

License Type	Error	Solution
KMS	Cannot contact KMS server	Verify network connectivity, check KMS server address
MAK	Activation limit reached	Contact Microsoft support for reset
Retail	Hardware changed too much	Use phone activation
OEM	OEM key not valid	Contact Microsoft or use new license

Domain Rejoin Failures

Symptom: Domain rejoin fails with error

Common errors:

1. "The specified domain either does not exist or could not be contacted"
- Check DNS configuration

◦ Verify network connectivity to domain controllers

◦ Test: `nslookup example.com`

2. "The account already exists"

- Computer object still exists in AD
- Run AD cleanup script: `C:\hyper2kvm\activedirectory\ad-cleanup.ps1`

3. "Access denied"

- Incorrect domain admin credentials
- Verify credentials and retry

Performance Issues

Symptom: Poor disk or network performance

Verification:

```
# Check VirtIO drivers loaded
Get-WmiObject Win32_PnPSignedDriver | Where-Object {$_.DeviceName -like
    "*VirtIO*"}

# Verify MSI interrupts
C:\hyper2kvm\performance\msi-verify.ps1

# Check TRIM enabled
fsutil behavior query DisableDeleteNotification
```

Solutions: 1. Verify VirtIO drivers installed correctly 2. Enable MSI interrupts if not already enabled 3. For SSD storage, verify TRIM enabled 4. Check KVM host configuration (CPU pinning, huge pages)

Application Compatibility Issues

Symptom: Application fails to start or function incorrectly

Steps: 1. Review compatibility report: `cat compatibility-report.md` 2. For hardware-dependent apps: Contact vendor for license transfer 3. For license services: Verify license server connectivity 4. For SQL Server: Run reconfiguration script 5. For dongles: Configure USB passthrough in KVM

Event Viewer Errors

Common errors after migration:

Event ID	Source	Meaning	Solution
10016	DistributedCOM	DCOM permissions	Usually harmless, can be ignored
7031	Service Control Manager	Service crashed	Check specific service logs
1014	DNS Client	DNS resolution failed	Verify DNS settings
6005	EventLog	System started	Normal, indicates successful boot

Additional Resources

- [Windows Configuration Schema Reference](#)
- [Application Compatibility Schema](#)
- [Performance Optimization Schema](#)
- [VirtIO Drivers Documentation](#)
- [Advanced Windows Support Roadmap](#)

Support

For issues or questions:

1. **Check logs:** `/var/log/hyper2kvm/migration.log`
2. **Review documentation:** <https://github.com/hyper2kvm/docs>
3. **Report bugs:** <https://github.com/hyper2kvm/issues>

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